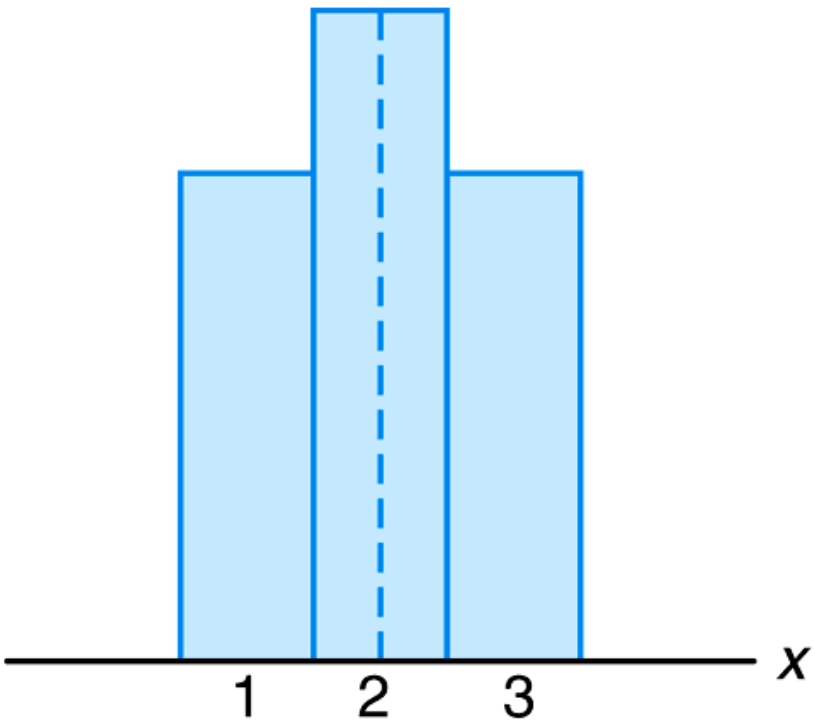
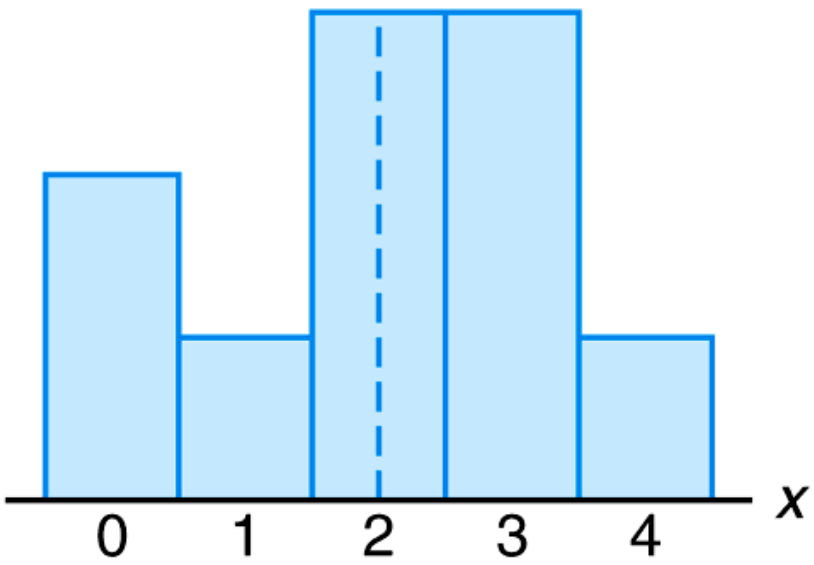
Variance & Standard Deviation

How can we mesuare the spread for a given data squence.

The variance of a random number is a measure of statistical dispersion.



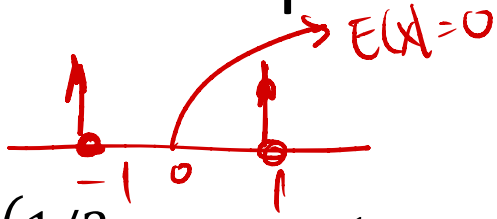
(a)



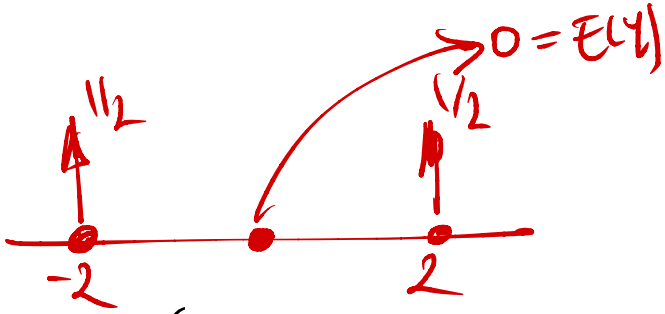
(b)

Instructive Example

Given



$$P_X(x) = \begin{cases} 1/2 & x = -1 \\ 1/2 & x = 1 \\ 0 & otherwise \end{cases}$$



$$P_Y(y) = \begin{cases} 1/2 & y = -2 \\ 1/2 & y = 2 \\ 0 & otherwise \end{cases}$$

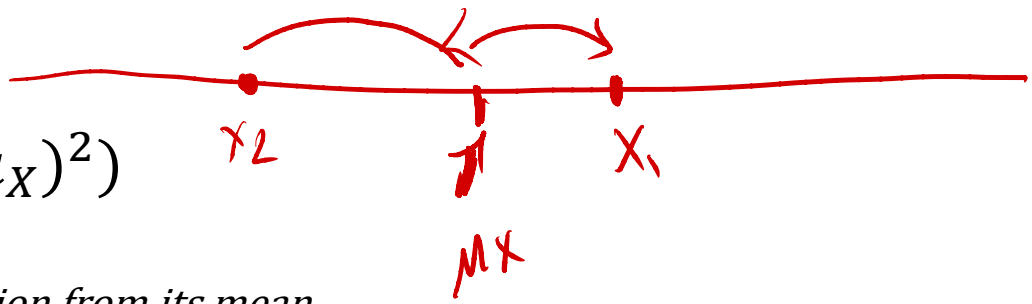
First Question: What are the expected values of X and Y ?

The expected value of X and Y equals to 0

Second Question: How much the random variables X and Y fluctuate about their mean ?

Definition: Variance and Standard Deviation

Definition: The variance of random variable X is expected value of g(x)=$(X-\mu_X)^2$



$$Var(X) = \sigma_X^2 = E((X-\mu_X)^2)$$

$X-\mu_X$ is called deviation of observation from its mean

Definition: The standard deviation of random variable X is

$$\sigma_X = \sqrt{Var(X)}$$

Instructive Example

$E((X-\mu_X)^2) = \sum_{-1,1} (x-\mu_X)^2 \cdot P_X(x) \Rightarrow 1^2 \cdot 1/2 + (-1)^2 \cdot 1/2 = 1$

$E((Y-\mu_Y)^2) = \sum_{-2,2} (y-\mu_Y)^2 P_Y(y) = 2^2 \cdot 1/2 + (-2)^2 \cdot 1/2 \Rightarrow 4$

Given

$$P_X(x) = \begin{cases} 1/2 & x = -1 \\ 1/2 & x = 1 \\ 0 & otherwise \end{cases} \qquad P_Y(y) = \begin{cases} 1/2 & y = -2 \\ 1/2 & y = 2 \\ 0 & otherwise \end{cases}$$

First Question: What is the expected value of X and Y ?

The expected value of X and Y equals to 0

Second Question: How much the random variable X and Y fluctuate about their mean ?

$$Var(Y) > Var(X)$$

Instructive Example

There is a chance that a bit transmitted through a digital transmission channel is received in error. Let X equal the number of bits in error in the next four bits transmitted. The possible values for X are {0,1,2,3}. The probability of making an error is 0.1.

$$E((X-\mu_X)^2) = \sum_{0,1,2,3,4} (x-\mu_X)^2 \cdot P_X(x)$$

$P(X = 0) = 0.6561$
$P(X = 1) = 0.2916$
$P(X = 2) = 0.0486$
$P(X = 3) = 0.0036$
$P(X = 4) = 0.0001$

$$E(X) = 0P(X = 0) + 1P(X = 1) + 2P(X = 2) + 3P(X = 3) + 4P(X = 4) = 0.4$$

x	$x - 0.4$	$(x - 0.4)^2$	$P(X = x)$	$(x - 0.4)^2 P(X = x)$
0	-0.4	0.16	0.6561	0.104976
1	0.6	0.36	0.2916	0.104976
2	1.6	2.56	0.0486	0.124416
3	2.6	6.76	0.0036	0.024336
4	3.6	12.96	0.0001	0.001296
				=0.36

Theorem 2.7:

For any random variable X (discr. or cont.)

$Var(X) = \sigma_X^2 = E((X - \mu_X)^2)$

$$\sigma_X^2 = E(X^2) - \mu_x^2$$

<u>Proof:</u>

$$Var(X) = \sigma_X^2 = E((X-\mu_X)^2) \Rightarrow$$

$$= E(X^2 + \mu_X^2 - 2X\mu_X)$$

$$= E(X^2) + E(\mu_X^2) - E(2X\mu_X)$$

$$= E(X^2) + \mu_X^2 - 2\mu_X \underbrace{E(X)}_{\mu_X}$$

$$= E(X^2) + \mu_X^2 - 2\mu_X^2$$

$$= E(X^2) - \mu_X^2$$

Instructive Example

There is a chance that a bit transmitted through a digital transmission channel is received in error. Let X equal the number of bits in error in the next four bits transmitted. The possible values for X are {0,1,2,3}. The probability of making an error is 0.1.

$P(X = 0) = 0.6561$
$P(X = 1) = 0.2916$
$P(X = 2) = 0.0486$
$P(X = 3) = 0.0036$
$P(X = 4) = 0.0001$

$$E(X) = 0P(X = 0) + 1P(X = 1) + 2P(X = 2) + 3P(X = 3) + 4P(X = 4) = 0.4$$

x	$(x)^2$	$P(X = x)$	$(x)^2 P(X = x)$
0	0	0.6561	0
1	1	0.2916	0.2916
2	4	0.0486	0.1944
3	9	0.0036	0.0324
4	16	0.0001	0.0016
			=0.52-0.16=0.36

Variance of a Function

Theorem 2.8

For any random variable X (discr. or cont.)

$$Var[aX \pm b] = a^2 Var[X]$$

<u>Proof:</u>

$y = ax - b$

$$Var(y) = E(y^2) - E(y)^2 = E((ax-b)^2) - (E(ax-b))^2$$
$$= E(a^2x^2 + b^2 - 2abx) - (E(ax) - E(b))^2$$
$$= E(a^2x^2) + E(b^2) - 2abE(x) - (E(ax)^2 + E(b)^2 - 2E(ax)E(b))$$
$$= a^2E(x^2) + b^2 - 2ab\mu_x - (E(ax)^2 + b^2 - 2ab\mu_x)$$
$$= a^2E(x^2) + \cancel{b^2} - \cancel{2ab\mu_x} - a^2E(x)^2 - \cancel{b^2} + \cancel{2ab\mu_x}$$
$$= a^2E(x^2) - a^2E(x)^2 \Rightarrow a^2\underbrace{(E(x^2) - E(x)^2)}_{Var(x)}$$
$$= a^2 Var(x)$$

Example 2.12

$E(X) = 0$

Given

$$P_X(x) = \begin{cases} 1/2 & x = -1 \\ 1/2 & x = 1 \\ 0 & otherwise \end{cases}$$

$Var(Y) = Var(3X-2)$

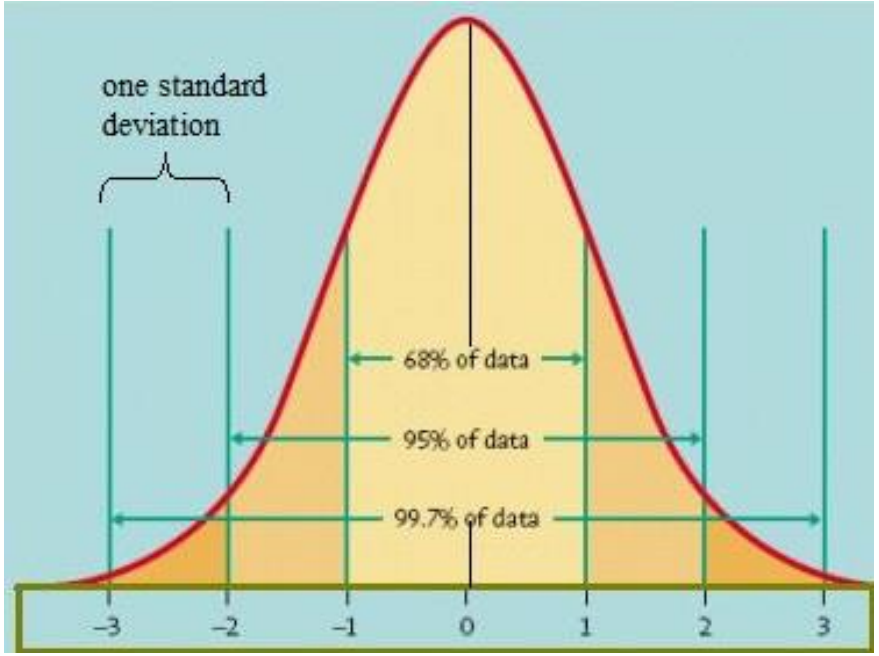
$3^2 Var(X) = 9$

$Var(X) = 1$

Find the variance of $Y = 3X - 2$

Physical significance of standard deviation

68-95-99.7 Rule



$$\Pr(\mu - 1\sigma \le X \le \mu + 1\sigma) \approx 68.27\%$$
$$\Pr(\mu - 2\sigma \le X \le \mu + 2\sigma) \approx 95.45\%$$
$$\Pr(\mu - 3\sigma \le X \le \mu + 3\sigma) \approx 99.73\%$$

One standard deviation above and below the mean is expected to include an observation with probability 68%.

Theorem 2.9: Chebyshev's Inequality

(Chebyshev's Theorem) The probability that any random variable X will assume a value within k standard deviations of the mean is at least $1 - 1/k^2$. That is,

$$P(\mu - k\sigma < X < \mu + k\sigma) \geq 1 - \frac{1}{k^2}.$$

$P(\mu - \sigma < X < \mu + \sigma)$

- Constraint on the value of k being $k > 0$ and real
- Lower limit on the value of probability of X
- Applicable to Cont. or Disc. R.V
- Distribution-free

Instructional Example: Chebyshev's Theorem

Suppose you know a dataset has a mean of 100 and a standard deviation of 10, and you're interested in a range of ± 2 standard deviations.

$\mu_X = 100$

$\sigma = 10$

$P(100 - 2.10 < X < 100 + 2.10)$

$P(80 < X < 120) \geq 1 - 1/2^2 = 3/4$

(Chebyshev's Theorem) The probability that any random variable X will assume a value within k standard deviations of the mean is at least $1 - 1/k^2$. That is,

$$P(\mu - k\sigma < X < \mu + k\sigma) \geq 1 - \frac{1}{k^2}.$$

Example - 2.13

In an experiment with three customers entering the Phonesmart store, the observation is N, the number of phones purchased. The PMF of N is

$$P_N(n) = \begin{cases} (4-n)/10 & n = 0, 1, 2, 3 \\ 0 & \text{otherwise.} \end{cases}$$

Find

a) The expected value, $E[N]$
b) $E[N^2]$
c) Variance, $\mathrm{Var}[N], \sigma^2$
d) Standard deviation, σ
e) $E((2+N)^2)$
f) $\mathrm{Var}(4+3N)$

Example - 2.13 cont.

$P_N(n) = (4-n)/10 \quad n = 0, 1, 2, 3$

a) $E(N) = 0 \cdot 4/10 + 1 \cdot 3/10 + 2 \cdot 2/10 + 3 \cdot 1/10 = 1$

b) $E(N^2) = 0^2 \, 4/10 + 1^2 \, 3/10 + 2^2 \, 2/10 + 3^2 \, 1/10 = 2$

c) $Var(N) = E(N^2) - \mu_n^2 = 2 - 1^2 = 1$

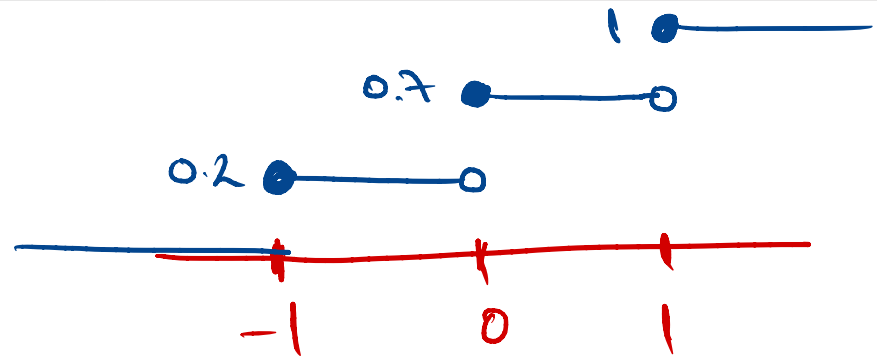
d) $\sigma_n = \sqrt{1} = 1$

e) $E((2+N)^2) \Rightarrow E(4 + N^2 + 4N) \Rightarrow E(4) + E(N^2) + E(4N)$

$= 4 + 2 + 4 \times 1 = 10$

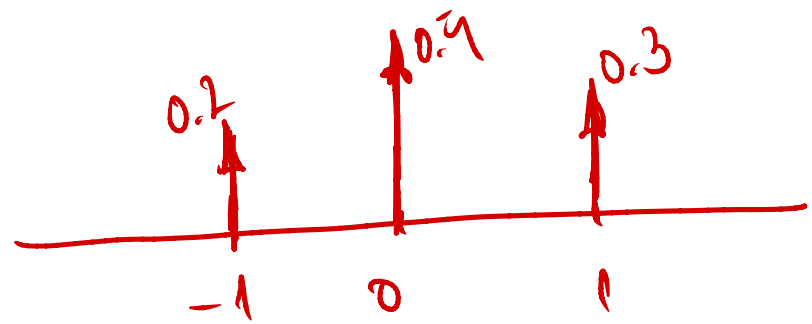
f) $Var(4+3N) \Rightarrow 3^2 Var(N) = 9 \times 1 = \underline{\underline{9}}$

Example - 2.14



Given

$$F_X(x) = \begin{cases} 0 & x < -1 \\ 0.2 & -1 \le x < 0 \\ 0.7 & 0 \le x < 1 \\ 1 & x \ge 1 \end{cases}$$



a) Draw a graph of CDF
b) Draw and write PMF
c) Find E[X],
d) Find Var[X]

$$E(X) = \sum_{-1,0,1} x \cdot P_X(x) = -1 \times 0.2 + 0 \times 0.5 + 1 \times 0.3$$

$$Var(X) = E(X^2) - \mu_X^2$$

$$E(X^2) = \sum_{-1,0,1} (-1^2) \times 0.2 + 0^2 \times 0.5 + 1^2 \times 0.3$$

Example - 2.14 cont.

Theorem 2.10 (variance for common distributions)

- The bernoulli random variable X has variance $\mathrm{Var}[X] = p(1-p)$
- The geometric random variable X has variance $\mathrm{Var}[X] = (1-p)/p^2$
- The binomial random variable X has variance $\mathrm{Var}[X] = np(1-p)$
- The poisson random variable X has variance $\mathrm{Var}[X] = \lambda T$
- The uniform variable X has variance $\mathrm{Var}[X] = (max - min)$(max-min+2)/12